

Saurabh Mandlik

Bengaluru, India | 8446223477 | mandliks1996@gmail.com | linkedin.com/in/saurabh-mandlik | github.com/mandu010

SUMMARY

Platform & Cloud Infrastructure Engineer with **8 years** automating enterprise infrastructure across **Azure and GCP** — currently governing an **850 PB multi-cloud storage estate** at Walmart. Deep in **Python automation**, distributed data pipelines, observability-driven reliability (Prometheus, Grafana, OpenTelemetry), and **FinOps cost optimization** (\$12M+ realized savings in H1 FY27). Targeting Senior/Staff Platform Engineering and Infrastructure roles.

EXPERIENCE

Software Engineer 3

Nov. 2023 – Present

Walmart Global Tech

Bengaluru, India

Multi-Cloud Storage KPI & Metadata Platform

- **Architected and solo-engineered** an enterprise multi-cloud automation platform providing a unified compliance framework across an **850 PB multi-cloud storage footprint** (150 PB Azure / 700 PB GCP) spanning **13K Azure storage accounts** and **300K+ GCP buckets**
- **Optimized Azure ingestion infrastructure** by re-engineering a legacy sequential PowerShell pipeline into a high-throughput concurrent Python engine (`concurrent.futures`), compressing metadata collection runtime **from 5 hours to 7 minutes (40x faster)**
- **Designed weekly automated BigQuery pipelines** to ingest telemetry across 300K+ GCP buckets (700 PB); engineered the framework to ingest scale workloads within provider API quota bounds to drive enterprise FinOps cost-optimization policies

MCP Server for Cloud Storage Infrastructure

- **Contributed to** a production-grade **Model Context Protocol (MCP) server** (FastMCP, FastAPI) exposing 8 tools for Azure Storage and BigQuery behind **PingFederate OAuth**; **owned the BigQuery tooling** — query execution and dataset retrieval — with async I/O, retry, deployed as stateless K8s service with **OpenTelemetry** tracing and **40+ Prometheus metrics**
- Leveraged the MCP server to tackle GCP **429 errors**; exported **1 year of daily egress/error metrics** to BigQuery, then used MCP in **Wibey (SuperAgent)** with **linear-deviation forecasting** to predict rate-limit breaches

Team-Based Deliverables

- Drove enterprise adoption of GCP **Autoclass** across a **600+ PB estate** — built the eligibility analysis engine and designed the phased opt-in rollout (non-prod → prod, opt-out governance); program realized **\$12M+ in savings in H1 FY27** with **\$29M+ additional identified**; currently evaluating Azure **smart-tiering** for org-wide adoption
- Contributed to TLS 1.0 deprecation planning and TRID tagging resolution, improving cost allocation accuracy

Cloud App Senior Automation Developer

Aug. 2021 – Nov. 2023

Accenture

Bengaluru, India

- **Windows Patching:** Built a 3-stage pipeline (Rundeck → Python → PowerShell) pulling server lists from **ServiceNow CRQ attachments** via API, creating **Azure Update Manager** schedules — reduced manual effort for **300 servers** from **15+ hours to a single CRQ input**, bi-weekly to monthly cadence
- **Linux Patching:** Automated weekly RPM patching for **25–30 servers** via Rundeck → Python → Ansible with pre/post-check diffs and consolidated email reports — cut effort from **20+ hours (2–3 engineers) to 1–2 hours (1 operator)**
- **Daily Inventory:** Scheduled PowerShell across **15 regional jump-servers** collecting agent inventory from VMware, Azure, and AWS; auto-delivered to supply chain — saved **400+ hours/month**
- **Decommission:** Built 2-phase pipeline — **soft decom** (stop/deallocate) then **hard decom** (delete VMs + resources) — automated **10+ servers per batch**, reducing **3–4 hours to minutes**
- **Server Provisioning:** Automated end-to-end: ServiceNow intake → VM creation → post-provisioning (domain join, agents, baseline) → operational readiness testing — reduced per-server turnaround from **2+ hours to a single automated run**

Sr Automation Analyst

Jul. 2018 – Aug. 2021

Capgemini

Mumbai, India

- **VM Build Automation** — Automated VM provisioning workflows using Python, PowerShell, and Micro Focus OO on VMware
- **vCenter & ESXi Patching** — Automated vCenter and ESXi host patching using PowerShell and Micro Focus OO
- **Universal Health Check Bot** — Built a PowerShell bot performing checks for AD, DNS, DHCP, Printer, SQL, and VMware; deployed across multiple clients

PROJECTS

Automated Algorithmic Trading System | *Python, asyncio, PostgreSQL, WebSockets, Slack* Jan. 2025 – Present

- **Architected and scaled** a fully autonomous multi-instrument trading platform over a 6+ year lifecycle, executing across **4 production environments** managing a live **INR 20L capital portfolio** with **100% unattended, zero-intervention runtime** for over 1 year
- **Engineered an async event-driven framework** utilizing **asyncio**, real-time WebSocket streaming, and non-blocking **asyncpg** connection pooling across 40+ modules to parallelize market data fetches and minimize execution slippage
- **Programmed deterministic risk guardrails and operations** by coding an ATR-driven 3-tier trailing stop-loss state machine, protected by automated position-sizing boundaries and self-healing bash lifecycle scripts with real-time Slack webhooks

TECHNICAL SKILLS

Languages: Python, PowerShell, Shell/Bash, SQL

Cloud & Infrastructure: Azure, GCP, AWS, Linux, Kubernetes, Docker, VMware, BigQuery, Digital Ocean

Automation & IaC: Terraform, Ansible, FastAPI, FastMCP, Pydantic, asyncio, Rundeck

Observability & DevOps: Prometheus, Grafana, OpenTelemetry, CI/CD (Jenkins, Azure DevOps), Git, GitOps, ServiceNow, REST APIs

CERTIFICATIONS

Azure Fundamentals (AZ-900) | HashiCorp Terraform Associate (003)

EDUCATION

GHRCE, RTMNU

B.E. — Electronics and Telecommunication Engineering

Nagpur, MH

Jun. 2018

ACHIEVEMENT

Dec 2022 — Led team to win “**Best Project**” in Accenture’s Intelligent Monitoring and Automation Bootcamp hosted by the Australian and French team.